MINI PROJECT PRESENTATION

PRODUCT REVIEW SENTIMENT ANALYSIS

Understanding customer opinions using Natural Language Processing and Machine Learning

Presented by:

Vishal Maurya | Roll No. 202410101150067
Akshat Kr Diwakar | Roll No. 202410101150080

Faculty: Mr Abhishek Sir

Subject / Project:
Product Review Sentiment Analysis

Introduction

What does sentiment analysis do?

What is the project?

It analyzes written product reviews and identifies the opinion expressed by the customer.

The output is a sentiment label: Positive, Negative or Neutral.

Why is it useful?

Online stores receive thousands of reviews. Reading each review manually takes time.

Automatic analysis helps summarize customer feedback quickly.

Example review

"The battery backup is excellent and the laptop works smoothly."

POSITIVE

Problem Statement & Objectives

The project turns unstructured text into useful feedback.

Problem statement

Customer reviews contain valuable opinions, but they are written in natural language.

Manually sorting reviews into positive, negative and neutral categories is slow and may be inconsistent.

A simple automated system is needed to classify reviews and summarize the results.

Project objectives

- Accept a product review as text
- Clean and prepare the review
- Convert text into numerical features
- Predict sentiment using a machine-learning model
- Display sentiment counts and percentages
- Present results in a simple dashboard

Expected outcome: a beginner-friendly application that makes review analysis easier.

Dataset & Input

Reviews are the main input for training and testing.

Dataset source

Amazon product reviews can be used as the dataset source.

For the basic demonstration, a small sample CSV file named reviews.csv can be created.

Dataset fields

review — the customer's written feedback
sentiment — the target label

Labels used: Positive, Negative and Neutral.

Illustrative sample records

Review text	Sentiment
Excellent quality and fast delivery	Positive
The product stopped working	Negative
The product is okay for its price	Neutral

Note: sample rows are illustrative; model performance must be measured after training.

Project Workflow

From raw customer text to a sentiment prediction.



Collect

Gather reviews in CSV format.

Clean

Remove noise and normalize text.

Vectorize

Convert text using TF-IDF.

Train

Train Logistic Regression.

Predict

Classify new reviews.

Display

Show results in Streamlit.

Main pipeline

reviews.csv → text preprocessing → TF-IDF features → Logistic Regression → sentiment result

Text Preprocessing & Feature Extraction

Machine-learning models need text in a numerical form.

1. Text preprocessing

- Convert text to lowercase
- Remove extra spaces and unwanted symbols
- Remove unnecessary noise
- Keep the cleaned review ready for analysis

Purpose: improve consistency in the input text.

2. TF-IDF vectorization

TF-IDF means Term Frequency– Inverse Document Frequency.

It converts words into numerical values based on how important they are in a review compared with the full dataset.

These numerical features are passed to the classifier.

Example: “ great battery life” → cleaned words → TF-IDF feature vector

Machine Learning Model

Logistic Regression is used for the basic classification task.

Why Logistic Regression?

- Suitable for text-classification tasks
- Simple and fast to train
- Works with numerical TF-IDF features
- Can classify a review into sentiment categories

The model learns patterns from labelled examples.

Training & evaluation

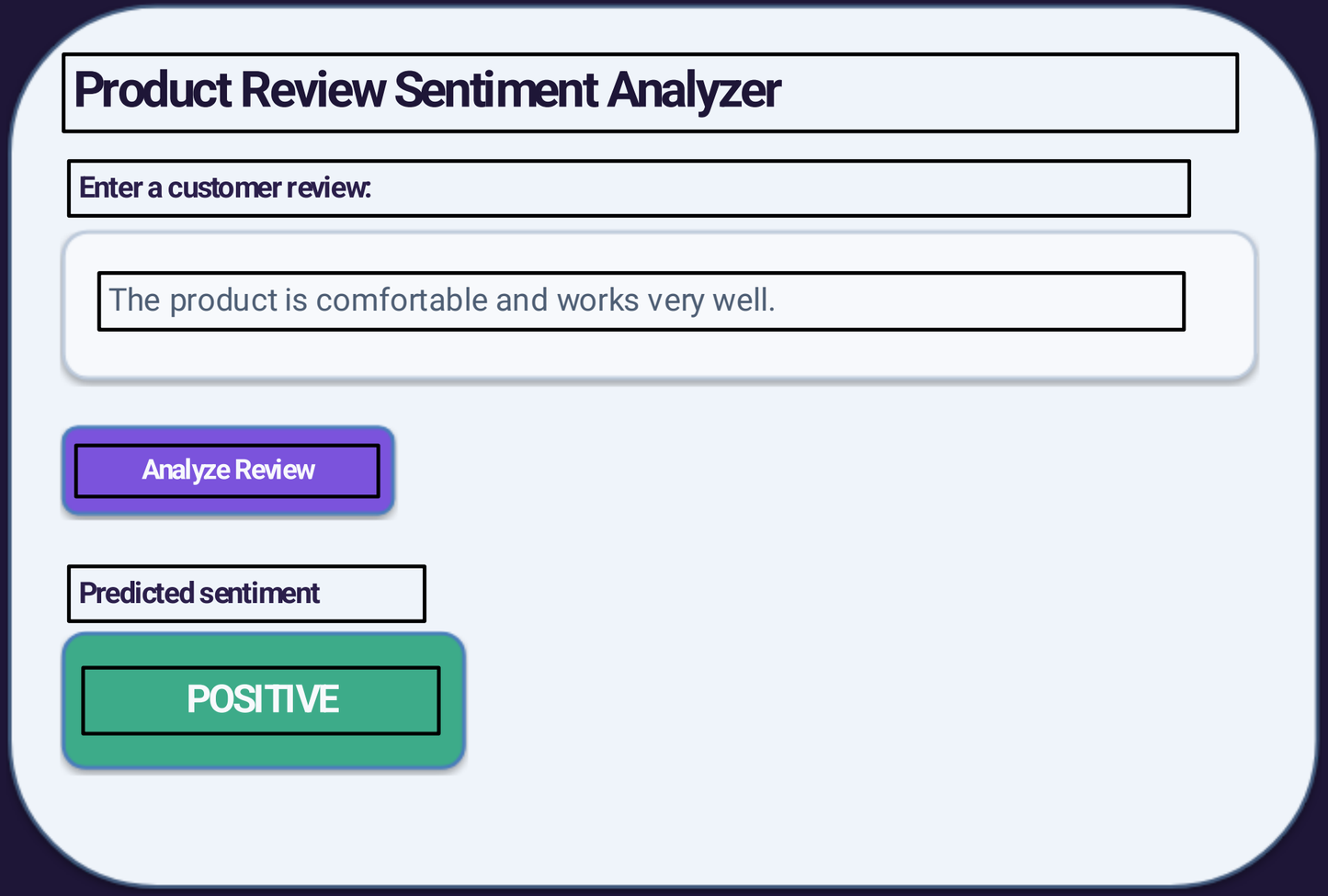
1. Split labelled data into training and testing sets.
2. Fit TF-IDF and Logistic Regression using training data.
3. Predict labels for test reviews.
4. Evaluate with accuracy, precision, recall and F1-score.

Report actual scores after running the model.

Important: no accuracy percentage is assumed in this presentation; it depends on the dataset and test results.

Application & Expected Output

A simple Streamlit interface makes the model easy to use.



Dashboard features

- Review input and prediction
- Positive / Negative / Neutral counts
- Sentiment percentages
- Charts for feedback overview
- Easy-to-understand output

Interface shown is a design mock-up. Actual results appear after running the application.

Conclusion & Future Scope

A basic NLP project with practical real-world applications.

Conclusion

Product Review Sentiment Analysis uses NLP and machine learning to understand customer opinions.

The system processes review text, converts it into TF-IDF features and predicts a sentiment label.

A Streamlit dashboard presents the prediction and overall feedback summary.

Future enhancements

- Use a larger and more diverse review dataset
- Improve accuracy through model comparison
- Support multiple product categories
- Add aspect-based analysis (battery, price, quality)
- Deploy the dashboard online

THANK YOU

Questions & Answers